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Breaking the Single Voice: Perceived AI Plurality as an Autonomy-Supportive Architecture for Ethical Marketplace Decisions

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Abstract

Artificial intelligence (AI) increasingly structures how consumers encounter, evaluate, and select marketplace options, raising fundamental questions about autonomy, empowerment, and welfare-enhancing choice environments. While prior work has focused on what algorithms recommend, less is known about how recommendation system architecture shapes the experience of choice. Across three experiments, this research examines the societal and psychological implications of perceived algorithmic plurality, defined as consumers' perception that recommendation authority is distributed across multiple distinct AI agents, each guided by distinct evaluative criteria. Results show that structuring ethical recommendations through a plural AI architecture enhances consumers' self–brand connection. Study 2 demonstrates that plural recommendations increase perceived autonomy, which strengthens self–brand connection and purchase intentions. Study 3 shows these effects are strongest among low-knowledge consumers, suggesting plurality can function as an equity-enhancing design feature. By shifting attention from algorithmic outputs to the socio-technical structure of AI nudges, this research advances understanding of consumer emancipation in digital marketplaces.

Keywords

algorithmic plurality, AI recommendations, nudge architecture, ethical marketing, consumer autonomy, socio-technical architectures, brand-self connection, welfare-enhancing design

Introduction

Artificial Intelligence (AI) has become a dominant institutional force in the marketplace, increasingly shaping not only what consumers see but how they choose. As digital platforms integrate AI into search, navigation, and product recommendation systems, the architecture of marketplace choice is no longer a neutral backdrop but a socio-technical infrastructure that structures agency, influences societal outcomes, and redistributes decision power between institutions and individuals (Gkikas & Theodoridis, 2021; Schwarz, 2025). Recommendation tools such as widely used Amazon's Choice badge exemplify this shift: by condensing complex algorithmic evaluations into a single authoritative cue, platforms evaluate AI from a decision aid to a centralised decision authority (Luo et al., 2026; Necula & Păvăloaia, 2023). In doing so, algorithmic systems do not simply facilitate choice; they govern it, carrying significant implications for market fairness, consumer autonomy, and broader patterns of ethical and sustainable consumption. These systems now shape an enormous share of marketplace activity, with Amazon attributing roughly 35% of its sales to algorithmic recommendations (MacKenzie et al., 2013) and 84% of e-commerce firms reporting AI integration as a strategic priority (Statista, 2024).

Moreover, prior research has largely conceptualized recommendation systems as singular decision authorities, despite the fact that platforms deploy multiple AI models simultaneously (Luo et al., 2026; Smith & Linden, 2017). Presenting recommendations through a unified AI voice (e.g., the Amazon's Choice badge) may improve efficiency (Luo et al., 2026), but this efficiency can also position the algorithm as the primary decision maker and increase consumer dependence (Bhardwaj et al., 2024; Bharti et al., 2026; Puntoni et al., 2020). Consequently, by delivering specific recommendations through a unified voice, these systems can act as negative nudges that

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discourage exploration and potentially reduce consumer agency. Furthermore, reduced consumer agency may also weaken consumers' relationships with brands. When algorithms guide choices with little deliberation, decision ownership can shift from the consumer to the system (Puntoni et al., 2020), diminishing brand–self connection (Bharti et al., 2026). Because brand–self connection reflects identity expression (Escalas & Bettman, 2003), reduced ownership risks turning consumption into a transactional act. Ethical consumption is especially identity-relevant, allowing consumers to express moral commitments and aspirational selves (Griskevicius et al., 2010; Wan & Bharti, 2025; White et al., 2019). If AI streamlines these choices, it may reduce opportunities for such identity expression. Because ethical decisions often involve perceived trade-offs, a sense of autonomy is essential for experiencing them as self-endorsed rather than externally imposed (Bandura, 2001; Deci & Ryan, 2000). A definitive or single AI recommendation can crowd out deliberation and position consumers as followers of algorithmic authority, reducing autonomy and weakening authorship over decisions with environmental consequences (Bharti et al., 2026; Puntoni et al., 2020).

This raises an important and underexplored question for ethical consumption contexts: can the design of AI nudging architecture, specifically, presenting AI recommendations as plural rather than singular, better support consumers in making empowered and socially meaningful choices? In addressing this question, it is important to recognize that recommendation systems are rarely monolithic. Platforms typically deploy multiple models optimized for distinct objectives, such as price efficiency, engagement likelihood, or sustainability metrics. For example, platforms such as Amazon rely on multiple AI models in the backend but present their outputs to consumers as a single recommendation (Adomavicius & Tuzhilin, 2005). More broadly, digital platforms translate complex algorithmic inputs into a unified decision cue. We propose that this design choice is not inevitable. Instead of presenting a unified AI decision authority, platforms could make perceived algorithmic plurality (AI plurality) visible by presenting distinct AI "voices," each guided by different evaluative criteria. We define perceived AI plurality as consumers' perception that recommendation authority is distributed across multiple distinct AI agents, each guided by distinct evaluative criteria, rather than concentrated in a single AI source. Importantly, this perception is conceptually distinct from simply displaying multiple AI sources or showing agreement among them; the central issue is not recommendation similarity, but whether consumers experience the recommendation environment as distributed rather than centralized.

Building on these gaps, this research makes four contributions to macromarketing. First, it reconceptualizes AI choice architecture and digital nudging as a form of market system design with societal consequences, showing that algorithmic plurality enhances consumer autonomy by counteracting the concentration of influence typical of unified recommendation systems. To the best of our knowledge, this is the first study in marketing to examine how plurality in AI nudging architecture shapes consumer autonomy and downstream brand responses. We show that presenting recommendations from multiple AI systems enhances perceived autonomy, shifting decision authority from algorithmic prescription to consumer authorship and strengthening self–brand connection and purchase intentions. Second, it shows that autonomy is pivotal in transforming ethical consumption from a compliance-driven act into an identity-relevant, self-endorsed decision (Chernev et al., 2011; Reed et al., 2012), deepening the societal meaning consumers derive from sustainable choices. Sustainable consumption often entails perceived sacrifice that limits adoption (Luchs et al., 2010; White et al., 2019), yet ethical choices can provide moral meaning and support identity expression (Griskevicius et al., 2010; Reed et al., 2012).

Third, this research contributes to consumer identity and self–brand connection literature by demonstrating that AI-mediated decision architectures can shape identity expression and relational outcomes, highlighting how autonomy-supportive AI design can foster more enduring consumer–brand relationships. Finally, adopting a macromarketing perspective on consumer welfare and marketplace inclusivity, we conceptualize AI recommendation architecture as part of the marketing system infrastructure through which consumers encounter and evaluate marketplace options (Layton, 2007; Mittelstaedt et al., 2006).We also show that the benefits of perceived AI plurality are not uniform. While prior research has shown that low consumer knowledge may hinder green consumption (Lee et al., 2020; Roh et al., 2022), our findings indicate that algorithmic plurality increases purchase intentions among less knowledgeable consumers, pointing to a pathway through which AI design can support more inclusive participation in ethical consumption. Taken together, across three randomized experiments, we examine how presenting ethical product recommendations through multiple AI systems shapes consumer responses. We address the following research questions:

RQ1: How does perceived AI plurality, defined as the presentation of recommendations through multiple AI agents (vs. a single unified AI voice), reshape consumers' self-brand connection in ethical consumption contexts?
RQ2: How does perceived AI plurality influence consumers' perceived decision autonomy in ethical consumption contexts?
RQ3: Does perceived autonomy mediate the effect of perceived AI plurality on self-brand connection and purchase intentions for ethical products?
RQ4: Under what conditions, such as consumer knowledge, does perceived AI plurality strengthen or weaken these effects?

Literature Review and Conceptual Development

AI Nudging and Algorithm Choice Architecture

The way choices are structured systematically shapes decision outcomes. Choice architecture refers to the design of decision

environments that influence behavior without restricting freedom of choice (Thaler & Sunstein, 2008; Thaler et al., 2012). Even subtle features such as defaults, ordering, and framing can alter preferences because there is no neutral way to present a choice (Johnson et al., 2012). By determining which options are visible and how they are presented, choice architects exert meaningful influence over decision processes. Building on this foundation, nudging emerges from this broader architecture as a behavioral design strategy that steers individuals toward particular outcomes while preserving choice freedom (Thaler & Sunstein, 2008). Because individuals rely on both automatic and reflective processes, subtle environmental cues can exert disproportionate influence, making choice architecture a powerful marketplace and policy instrument. Ethical debates, therefore, emphasize the responsibility of choice architects to balance behavioral influence with autonomy and welfare considerations (Selinger & Whyte, 2011).

As decision environments migrate online, digital environments have amplified the reach and precision of nudging. Digital nudging refers to the use of interface design elements and data-driven personalization to guide behavior in online environments (Weinmann et al., 2016). Technology-enabled decision aids reduce search costs and improve preference matching, yet they also shape choice by determining which attributes become salient and which alternatives are considered (Häubl & Murray, 2003; Häubl & Trifts, 2000). More recently, artificial intelligence has further expanded the influence of digital nudging by enabling predictive personalization. AI-driven recommendation systems analyze behavioral data to anticipate preferences and deliver targeted suggestions, reducing search costs and cognitive effort (Puntoni et al., 2020). Recommendation cues such as badges and rankings can improve decision efficiency and confidence under information overload (Kumari & Raj 2024; Luo et al., 2026). At the same time, algorithmic recommendations can function as heuristic signals of credibility, encouraging users to defer to guidance perceived as authoritative (Logg et al., 2019). While such reliance streamlines decisions, it may also shift consumers from active evaluators to passive recipients of algorithmic guidance.

Beyond facilitating choice, AI-enabled nudges shape psychological engagement with decisions. Recommendation transparency and preference match enhance decision confidence and perceived quality, whereas opaque or mismatched recommendations increase uncertainty and reduce confidence (Luo et al., 2026). Algorithmic guidance can also attenuate consumers' sense of accountability and authorship, as decision responsibility is attributed to the system rather than the self (Puntoni et al., 2020). Thus, AI-mediated nudges influence not only what consumers choose but how they experience the decision process.

Importantly, these psychological effects carry broader marketplace implications. When algorithmic cues are perceived as definitive, consumers may disengage from deliberation and rely more heavily on system guidance, narrowing exploration and reinforcing platform-driven demand patterns. Although such guidance enhances efficiency, opaque recommendation systems can reduce perceived control and autonomy, particularly when consumers do not understand how recommendations are generated (Dietvorst et al., 2015; Puntoni et al., 2020). Accordingly, scholars increasingly call for AI-enabled decision environments that preserve consumer engagement, accountability, and autonomy. Furthermore, most research conceptualizes recommendation systems as singular decision agents, even though platforms often rely on multiple models operating simultaneously. Consumers typically encounter outputs framed as a unified AI judgment, signaling that evaluative considerations have been resolved on their behalf. This framing positions the algorithm as a definitive decision cue rather than a decision-support tool, potentially diminishing deliberation and perceived autonomy.

Building on choice architecture theory, the present research introduces perceived AI plurality as a structural feature of algorithmic nudging. Rather than conveying a definitive solution, plural recommendations signal that value judgments remain open to personal interpretation. By reframing algorithmic guidance as a set of perspectives rather than a singular prescription, algorithmic plurality enhances perceived consumer autonomy by reinforcing the sense that the consumer authors the decision. From a macromarketing perspective, such design choices shape not only individual decisions but also patterns of marketplace participation, consumer capability, and demand formation. These dynamics are especially consequential in ethical and sustainable consumption contexts, where marketplace choices express personal values and identity commitments, making decision authorship essential for meaningful engagement with socially responsible options.

AI and Ethical Choices

Ethical products are goods produced with consideration for environmental sustainability, labor conditions, human welfare, and animal well-being (White et al., 2019). Despite growing concern for these issues, ethical consumption is constrained by structural barriers, including higher prices, perceived performance uncertainty, limited availability, and difficulty aligning purchases with specific moral priorities (Carrington et al., 2014; Casais & Faria, 2021). Consequently, ethical purchasing often requires navigating trade-offs between moral values and functional or economic considerations, contributing to the well-documented intention–behavior gap (Carrington et al., 2014).

Consumers' willingness to accept such trade-offs depends in part on the personal significance of the decision. Ethical consumption frequently requires prioritizing collective welfare over immediate self-interest, making the self's role central to whether ethical intentions translate into action (Carrington et al., 2014; White et al., 2019). Because conventional purchases rarely require comparable sacrifice, ethical consumption involves greater moral deliberation and personal responsibility.

Although research has advanced understanding of ethical consumption barriers and decision processes, this work has only recently intersected with research on AI-mediated choice

environments. Prior studies examine how service technologies influence ethical consumption (Choi & Yi, 2025), how AI marketing applications shape sustainability outcomes (Hermann, 2022), and how transparency and personalization affect trust in AI recommendations (Qadri et al., 2026). At the same time, research shows that AI recommendations can shift perceived responsibility away from the consumer, reducing choice accountability and decision engagement (Bharti et al., 2026; Puntoni et al., 2020). When responsibility is externalized to technological systems, consumers may disengage from value-based deliberation, weakening the translation of ethical intentions into behavior.

Because ethical consumption often entails perceived sacrifice and moral evaluation, decisions must be experienced as self-endorsed rather than externally imposed for ethical intentions to translate into behavior (Deci & Ryan, 2000; White et al., 2019). Decision environments that diminish personal authorship may therefore weaken engagement with ethical choices, whereas environments that preserve consumer agency can strengthen commitment to value-consistent behavior. Beyond behavioral outcomes, ethical consumption also carries symbolic and identity-relevant meaning. Choosing sustainable and socially responsible products allows consumers to express moral values and reinforce their sense of self (Belk, 1988; Reed et al., 2012). When decisions are experienced as personally meaningful, consumption can strengthen connections between the self and chosen brands, deepening relational bonds and reinforcing identity expression (Escalas & Bettman, 2005). From a macromarketing perspective, these dynamics are consequential because marketplace structures shape not only individual decisions but also patterns of ethical engagement and societal well-being. These identity-relevant outcomes suggest that ethical consumption contexts may strengthen self–brand connections, particularly when consumers experience decisions as personally meaningful and self-endorsed.

Brand-Self Connection

Brand–self connection refers to the extent to which individuals perceive a brand as linked to their sense of self, such that brand-related meanings become integrated with personal identity (Escalas, 2004). This connection arises when consumers see a brand as symbolically resonant, self-expressive, or reflective of their values, goals, and autobiographical memories (Moore & Homer, 2008). As a relational construct, brand–self connection captures the emotional and cognitive bond that forms when consumers use brands as instruments of identity construction. It is therefore foundational to understanding the broader role that brands play as cultural and psychological resources within marketplace systems.

Decades of research show that brand–self connection is a powerful predictor of consumer behavior. Strong self-brand ties are associated with heightened loyalty, increased willingness to pay, engagement with brand communities, and resilience to negative information (Cheng et al., 2011). Consumers who feel connected to a brand tend to interpret brand actions through a relational lens, respond more favorably to brand communications, and maintain supportive attitudes even in the face of failure. These outcomes arise because brand–self connections are not merely attitudinal preferences but are deeply rooted in self-definition and identity articulation. When a brand becomes part of the self, consumption choices take on greater psychological significance, reinforcing the coherence and continuity of one's identity.

A growing body of work suggests that brand–self connection is shaped not only by brand communications and cultural meanings but also by the structure of the decision environment. Emerging research on AI-mediated consumption shows that overreliance on algorithmic recommendations can weaken the psychological mechanisms that support brand–self ties leading to diminished authorship over their decisions (Bharti et al., 2026). This displacement of agency can undermine the sense of ownership and personal relevance necessary for sustaining strong brand connections. In other words, when algorithms rather than consumers drive the choice process, brand-related meanings become less personally constructed and less identity expressive.

This concern is particularly salient in ethical consumption contexts, where brands enable consumers to express moral commitments and aspirational identities (Griskevicius et al., 2010; White et al., 2019). If AI systems overly streamline such choices, consumers may have fewer opportunities to construct identity-relevant meaning. Conversely, decision environments that invite reflection and personal judgment may strengthen the integration of ethical brands into the self-concept. Presenting recommendations from multiple AI systems may foster such engagement. Rather than signaling a single authoritative solution, plural recommendations encourage consumers to interpret alternatives and select options aligned with their values. By reinforcing the perception that the decision reflects the self, plural recommendation architectures may strengthen self–brand connection.

In the present context, this logic applies to the specific AI-recommended ethical brands presented in the choice task. Because ethical and sustainable brands carry symbolic meanings that consumers can use to express values, moral commitments, and desired identities, a recommendation architecture that preserves perceived autonomy may strengthen consumers' connection with the recommended brands (Escalas & Bettman, 2003, 2005; McCracken, 1986; White & Dahl, 2007). Thus, when plural AI recommendations support decision authorship, consumers should be more likely to experience the recommended ethical brands as reflective of the self. Accordingly, we propose:

H1: A plural AI nudge architecture (vs. a singular AI architecture) enhances consumers' self–brand connection with the chosen ethical product.

Perceived Autonomy

Autonomy is widely regarded as a central dimension of human agency and consumer well-being. It reflects the capacity to

make choices that are personally meaningful, value-congruent, and self-directed (André et al., 2018; Cho and Rha, 2026). In marketplace settings, autonomy enables consumers to operate as intentional agents rather than passive recipients of external influence. Scholars have long argued that autonomy is foundational both normatively and empirically: it is essential for constructing an authentic sense of self, for enacting personal goals, and for experiencing consumption as meaningful rather than imposed. Recent scholarship formalizes autonomy as a market condition shaped by institutional structures, technological systems, and the distribution of decision power, emphasizing that true autonomy requires both freedom from external constraints and freedom to act in ways aligned with one's values (Hyman et al., 2022).

Contemporary digital marketplaces complicate these conditions. AI systems increasingly curate information, filter alternatives, and prioritize specific options, effectively shaping the boundaries of consumer choice. Although these systems reduce cognitive effort, they can also weaken perceived authorship by relocating evaluative responsibility from consumers to algorithmic intermediaries (André et al., 2018). When individuals attribute their decisions to an algorithm rather than to their own deliberation, they experience diminished perceived autonomy. This shift is psychologically consequential because autonomy is tied to authenticity, responsibility, and subjective well-being.

Self-determination theory identifies autonomy as a fundamental psychological need that supports motivation and volitional engagement (Deci & Ryan, 2000). Autonomy-supportive environments enhance personal endorsement and commitment, while constraints undermine satisfaction. Within digital consumption, autonomy supportive conditions include access to diverse alternatives, transparent evaluative information, and the ability to understand and navigate recommendation criteria (Nokhiz and Ruwanpathirana, 2025). These conditions are increasingly recognized as essential for maintaining consumer agency and autonomy in technology mediated contexts. Perceived autonomy carries important behavioral and economic implications. First, greater autonomy enhances satisfaction with consumption experiences, which in turn predicts willingness to pay and loyalty (Homburg et al., 2005). Similarly, research in service contexts shows that autonomy strengthens motivation and value internalization, improving relational outcomes (Auh et al., 2011). Together, these findings position autonomy as a psychological resource that creates marketplace value. Furthermore, autonomy is especially important in ethical consumption, where decisions involve identity expression, and perceived sacrifice, and where choices gain meaning when experienced as self-endorsed rather than externally directed (Carrington et al., 2014; White et al., 2019). When consumers feel responsible for resolving ethical considerations, sustainable decisions are more likely to be experienced as expressions of personal values and identity commitments.

Taken together, prior research portrays autonomy as both a personal experience and a structural condition shaping meaningful consumption. As AI increasingly structures decision environments, perceived autonomy becomes a critical mechanism through which AI choice architectures influence ethical decision making and identity-relevant responses. Accordingly, the influence of AI choice architecture on purchase intentions may operate through a psychological pathway in which perceived autonomy enhances self–brand connection, thereby increasing consumers' motivation to act on their preferences. Perceived autonomy should strengthen self-brand connection because it allows consumers to experience the final choice as self-authored rather than algorithmically imposed. This sense of decision authorship is especially important in ethical consumption contexts, where choices carry moral and identity-relevant meaning, enabling consumers to experience the recommended ethical brands as reflecting their own values and desired self-concept. In this view, autonomy and self–brand connection function as sequential mechanisms through which plural AI nudge architectures shape ethical product choice. Accordingly, we propose:

H2: A plural AI nudge architecture (vs. a singular AI architecture) enhances perceived choice autonomy.

H3: Perceived autonomy and self–brand connection sequentially mediate the effect of AI nudge architecture (plural vs. singular) on purchase intentions.

Consumer Knowledge

Consumer knowledge has long been recognized as a central determinant of marketplace decision making. Early work conceptualized consumer knowledge as both the amount and structure of information available to the individual, shaping how consumers interpret product attributes, evaluate alternatives, and enact choice (Oxenfeldt, 1950). Subsequent research formalized consumer knowledge as a multidimensional construct encompassing factual information, experience based familiarity, procedural understanding, and the ability to discriminate among product related cues (Ratchford, 2001). In this tradition, knowledge is not merely an index of stored information, but a resource that influences the efficiency, depth, and quality of consumer judgments.

Building on this foundation, contemporary research emphasizes that consumer knowledge varies substantially across individuals and product domains, producing marked differences in decision strategies. Pillai et al. (2015) demonstrate that knowledge disparities shape how consumers categorize information, attend to diagnostic cues, and discriminate between meaningful and irrelevant attributes. High knowledge consumers are better able to detect misleading or incomplete information, whereas low knowledge consumers rely more heavily on superficial cues, defaults, or externally provided recommendations. These findings reinforce a broader view of consumer knowledge as a structural condition that shapes not only cognitive processes but also vulnerability within marketplace systems.

In ethical consumption contexts, consumer knowledge plays an especially critical role because decisions often require the integration of complex information, value considerations, and

perceived tradeoffs. Research shows that knowledge enables consumers to evaluate ethical claims, assess product level consequences, and reconcile competing priorities (Bharti & Bharti, 2025). When knowledge is limited, consumers engage in less systematic information processing and depend more on salient cues that reduce cognitive effort. Such reliance increases susceptibility to primacy effects, framing influences, and persuasive messaging, leading to decisions that may be less aligned with personal values or long term goals (Bharti & Bharti, 2025). These dynamics are particularly important in contexts where ethical choices carry social or environmental implications, as limited knowledge can diminish perceived agency and reduce the meaningfulness of the choice process.

At the same time, high knowledge consumers tend to approach ethical decisions with greater deliberation. They possess a more elaborate schema for evaluating ethical attributes, can identify subtle tradeoffs, and engage in deeper processing of sustainability related information (Bharti & Bharti, 2025). This depth of evaluation can heighten perceived autonomy by reinforcing the sense that choices reflect personal deliberation rather than external cues. Knowledgeable consumers are therefore more likely to experience ethical decisions as self endorsed acts, strengthening the psychological significance of their choices and enhancing downstream outcomes such as satisfaction, identity alignment, and sustainable consumption intentions.

In the context of AI shaped decision environments, these knowledge based distinctions become particularly salient. When recommendation systems present cues that appear authoritative or definitive, low knowledge consumers may be more prone to defer to algorithmic guidance, thereby relinquishing some degree of perceived control. Conversely, knowledgeable consumers are better equipped to interrogate recommendations, compare alternatives, and maintain a sense of volitional agency. This divergence suggests that consumer knowledge functions as a structural boundary condition that shapes how individuals experience autonomy and meaning in AI mediated choice environments.

Taken together, literature positions consumer knowledge as a central factor that influences cognitive effort, susceptibility to external cues, and the authenticity of decision-making. Its role is especially pronounced in ethical consumption, where knowledge supports value consistent choices and reinforces the autonomy necessary for meaningful and identity-relevant marketplace behaviors. This suggests that not all consumers benefit equally from autonomy supportive AI nudging strategies, and the effectiveness of such architectures may depend on consumers' underlying knowledge resources. Accordingly, we propose the following moderated mediation hypothesis:

H4: Consumer knowledge moderates the effect of AI nudge architecture on perceived autonomy such that the positive effect of a plural (vs. singular) AI architecture on purchase intentions of the ethical product is stronger among consumers with lower levels of knowledge.

Methodology

Overview of Studies

Across three experiments, we examine how AI nudge architecture shapes ethical consumption outcomes. Study 1 tests the foundational effect of algorithmic plurality on consumers' self–brand connection (H1). Study 2 investigates the underlying mechanism by testing a serial mediation process in which perceived autonomy and self–brand connection sequentially transmit the effect of plural AI architecture on purchase intentions (H2, H3). Study 3 identifies a theoretically meaningful boundary condition by examining consumer knowledge as a moderator of the autonomy pathway and its downstream consequences for purchase intentions (H4). Across the three experimental studies, participants completed scenario-based online shopping tasks involving ethically positioned products, with product categories varied across studies to enhance external validity. Unless otherwise noted, all multi-item measures used nine-point scales (1 = strongly disagree, 9 = strongly agree). Sample items are reported below, and full scale details appear in Appendix A.

Study 1

Participants, Design, and Procedure. Study 1 examined whether presenting ethical recommendations through multiple AI agents enhances consumers' connection with the recommended brand relative to a single unified AI system. Participants were recruited through Prolific Academic (native English speakers only) and received monetary compensation. After excluding incomplete responses and failed attention checks, the final sample consisted of 212 participants (52.8% female; $M_{age} = 45.08$ years, $SD_{age} = 13.8$). The study employed a between-subjects design in which participants were randomly assigned to one of two AI nudge architectures: a singular AI recommendation system or a plural AI system consisting of three independent AI agents. Consistent with prior research on choice architecture (Iyengar & Lepper, 2000), we implemented AI plurality using a three-option structure, which introduces multiplicity while maintaining cognitive simplicity.

Participants were asked to imagine shopping online on a reputable retail website to purchase ethically produced coffee for home use. They viewed three coffee options positioned as ethical choices, each including brand information, ethical sourcing claims, product descriptions, and price information. The recommendation display indicated whether the suggested option was generated by one unified AI model or by multiple independent AI systems. In the plural-AI condition, participants were informed that the website used three independent AI models that applied distinct ethical priorities. The exemplar stimulus is presented in Figure 2.

After selecting the product they were most likely to purchase, participants completed a measure of self–brand connection (Escalas & Bettman, 2017; Cronbach's $\alpha = .80$). A sample

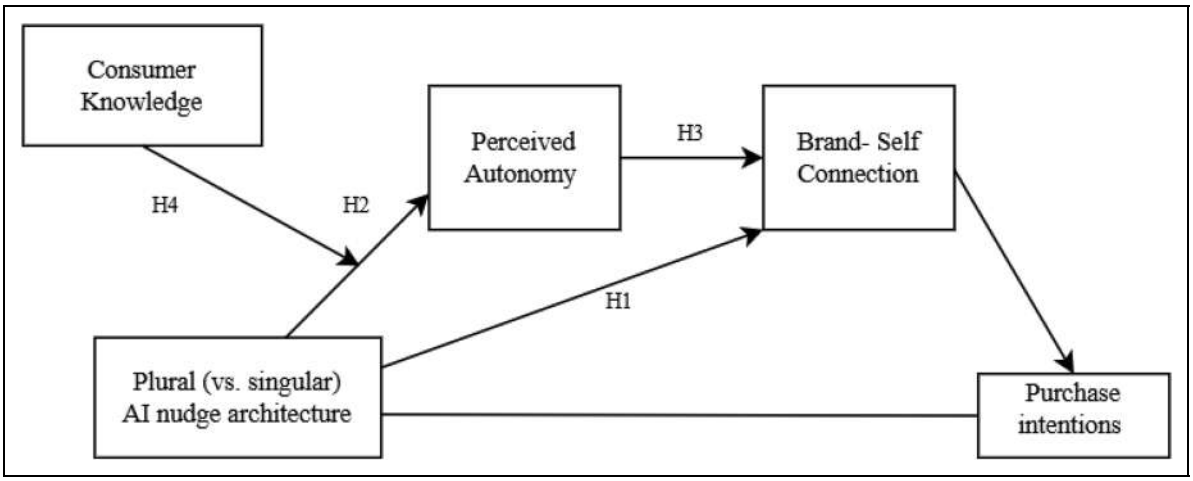


Figure 1. Conceptual framework.

item is "I can identify with the brands recommended by the platform's AI." The full scale is reported in Appendix A. Control variables included trust in AI recommendations, perceived task complexity, and attitudes toward ethical purchasing. A manipulation check verified participants' perception of the recommendation architecture. This study tests H1, which predicts that a plural AI nudge architecture enhances consumers' self–brand connection with the chosen product.

Results. Participants in the plural-AI architecture condition indicated a significantly higher mean for the manipulation check statement than participants in the single-AI architecture condition ($M_{plural\text{-}AI} = 6.9$, $SD = 2.07$ vs. $M_{single\text{-}AI} = 1.43$, $SD = 1.24$, $t(210) = 23.08$, $p < .001$). This confirms that plurality-based AI manipulation was perceived as intended.

A regression of self-brand connection on AI architecture (1 = plural AI, 0 = single AI), was conducted along with control variables, such as trust in AI recommendations, perceived task complexity, attitude toward ethical purchasing, and demographic factors. The results revealed a significant effect of AI architecture on self-brand connection ($\beta = 1.07$, $SE = .24$, $t(205) = 4.41$, $p < .001$), after controlling for relevant variables. Specifically, participants exhibited higher self-brand connection under the plural-AI architecture compared to the single-AI architecture ($Mean_{plural\text{-}AI} = 4.85$, $S.D. = 2.1$ vs. $Mean_{single\text{-}AI} = 3.78$, $S.D. = 1.8$, $t(210) = 3.96$, $p < .001$, Cohen's $d = .55$). Thus, H1 was supported.

Discussion for Study 1. Study 1 provides initial evidence that breaking the single algorithmic voice can meaningfully shape consumer–brand relationships. Compared to a unified AI nudge, a plural AI nudge architecture significantly enhanced self-brand connection, demonstrating that the structure of AI recommendations, beyond their content, can strengthen consumers' relational engagement with marketplace offerings.

Study 2

Participants, Design, and Procedure. Study 2 examined the role of perceived AI plurality on consumer choice autonomy (H2). It also sought to examine the psychological mechanism through which AI nudge architecture shapes ethical decision-making (H3). The design was similar to Study 1, except that a different product category was used and additional variables were measured to test the proposed serial mediation process. Specifically, we examined whether algorithmic plurality (multiple AI vs. single unified AI architecture) enhances self-brand connection through perceived autonomy and whether this carries meaningful downstream consequences, such as a higher ethical purchase intention.

Participants were recruited from Prolific Academic ($N = 400$; 53.8% female; $Ma_{ge} = 44.3$ years, $SD = 14.15$; native English speakers) and compensated for their participation. Similar to study 1, participants were randomly assigned to the two AI nudge architecture conditions (plural vs. singular). They imagined shopping online for ethically produced chocolate and viewed three options labeled as ethical, each including brand information, ethical sourcing claims, product descriptions, and prices. The recommendation label indicated whether the suggestion resulted from a single AI model or multiple independent AI systems emphasizing different ethical priorities (stimuli detailed in Appendix B). After selecting a product, participants completed measures of perceived autonomy (Cho & Rha, 2026; e.g., "The AI recommendation(s) allowed me to make my own decisions"), self–brand connection (Escalas & Bettman, 2003), and purchase intentions (Grewal et al., 1998; e.g., "The likelihood that I would select one of the ethical chocolate options shown is…"). Control variables mirrored those of Study 1 and included trust in AI recommendations, perceived task complexity, and attitudes toward ethical purchasing (full scales in Appendix A). This design enabled a test of H3, which predicts that perceived autonomy and self–brand connection sequentially mediate the effect of AI nudge architecture on purchase intentions.



Figure 2. Exemplar stimuli for plural AI nudge architecture (study 1). The detailed stimuli are presented in Appendix B.

Results. Participants in the plural-AI architecture condition reported significantly higher scores on the manipulation check assessing perceived algorithmic plurality than those in the single-AI architecture condition ($M_{plural\text{-}AI} = 7.18$, $SD = 2.16$ vs. $M_{single\text{-}AI} = 1.85$, $SD = 1.82$, $t(398) = 26.73$, $p < .001$), confirming that the manipulation was perceived as intended.

Participants in the plural-AI condition reported higher perceived autonomy than those in the single AI condition ($M_{plural\text{-}AI} = 6.32$, $SD = 1.68$ vs. $M_{single\text{-}AI} = 4.9$, $SD = 2.03$, $t(398) = 7.76$, $p < .001$, Cohen's $d = .76$), providing initial support for the proposed hypothesis H2. Similarly, participants in the plural-AI condition reported higher self-brand connection ($M_{plural\text{-}AI} = 4.61$, $SD = 2.1$ vs. $M_{single\text{-}AI} = 3.84$, $SD = 2.05$, $t(398) = 3.61$, $p < .001$, Cohen's $d = .36$), and purchase intentions ($M_{plural\text{-}AI} = 5.51$, $SD = 2.17$ vs. $M_{single\text{-}AI} = 4.9$, $SD = 2.14$, $t(398) = 2.36$, $p = .018$, Cohen's $d = .24$).

We tested the proposed sequential mediation H3 using PROCESS Model 6 with 5,000 bootstrap samples. AI nudge architecture (0 = unified AI, 1 = plural AI) was specified as the independent variable, perceived autonomy (PA) as the first mediator, self-brand connection (SBC) as the second mediator, and purchase intention (PI) as the dependent variable. Attitude toward ethical purchase, trust in AI recommendations, perceived task complexity, and demographic variables were included as covariates.

Consistent with our theorizing, perceived AI plurality significantly increased perceived autonomy ($b = 1.67$, $SE = .16$, $p < .001$), indicating that distributing recommendations across multiple AI agents enhanced consumers' sense of decision authorship. Perceived autonomy, in turn, positively predicted brand self-connection ($b = .22$, $SE = .06$, $p < .001$). Further, brand self-connection strongly predicted purchase intention ($b = .39$, $SE = .05$, $p < .001$). The direct effect of algorithmic plurality on purchase intention was not significant ($b = .17$, $SE = .19$, $p = .38$), suggesting full mediation.

Bootstrap analyses revealed a significant total indirect effect of perceived AI plurality on purchase intention (Effect = .51, 95% CI [.25, .79]). Further, the indirect effect through brand self-connection was significant (Effect = .2, 95% CI [.04, .38]). Finally, the serial indirect effect through perceived autonomy and brand self-connection was also significant (Effect = .14, 95% CI [.06, .24]). These findings support the proposed sequential mediation H3.

The full results are presented in Tables 1 and 2.

Discussion for Study 2. Results of study 2 reveal that autonomy's influence on purchase intentions operated through deeper self–brand connection. This pattern suggests that the welfare benefits of plural AI architectures emerge not simply because consumers feel more autonomous, but because restored authorship enables consumers to internalize the decision as self-expressive. Together, these findings suggest that plurality-based AI nudging enhances downstream behavioral intentions by strengthening consumer autonomy and fostering deeper self-brand connections in their ethical choice decisions.

Study 3

Participants, Design, and Procedure. Study 3 examined consumer knowledge as a boundary condition influencing the effectiveness of plural AI architecture (H4). Participants were recruited from Prolific Academic and compensated for their participation ($N = 510$). After screening procedures, the final sample consisted of 506 respondents (50.8% female; $M_{age} = 43.9$ years, $SD = 13.72$; native English speakers). Participants were randomly assigned to either plural or singular AI recommendation architecture and asked to imagine purchasing ethically produced toothpaste. They viewed three options positioned as ethical choices, each including ethical production claims, brand information, and pricing details. The recommendation display

Table 1. Regression Results for Study 2.

Outcome Variable:	Perceived Autonomy (PA)	Self-Brand Connection (SBC)	Purchase Intentions (PI)
constant	1.95	−.005	−.004
AI architecture (1 = plural)	1.65***	.53**	.18
	(.164)	(.2)	(.19)
PA		.22***	.1*
		(.06)	(.05)
SBC			.38***
			(.05)
PI			
Trust	.42***	.35***	.26***
	(.04)	(.05)	(.05)
att	.07	.15**	.11*
	(.04)	(.5)	(.04)
task comp	−.06	.12**	.01
	(.04)	(.05)	(.04)
gender	.29	−.14	−.07
	(.157)	(.17)	(.16)
age	−.004	−.009	.13*
	(.005)	(.006)	(.006)
income	−.05	−.009	.07
	(.06)	(.07)	(.07)

*$p < .05$; ** $p < .01$; ***$p < .001$; standard error (S.E.) in parenthesis; PA: Perceived autonomy; PI: purchase intention; SBC: Self-brand connection; task comp: perceived task complexity; att: attitude toward ethical purchasing.

Table 2. Serial Mediation Results.

Indirect Effects of AI Plurality	Effect	LCC	UCC
AI plurality —> PI	.51	.25	.8
AI plurality —> SBC —> PI	.2	.04	.38
AI plurality —> PA —> SBC —> PI	.14	.06	.23

PA: Perceived autonomy; PI: purchase intention; SBC: Self-brand connection; 95% CI.

indicated whether the suggested option originated from a single AI model or multiple independent AI systems.

Participants then completed the perceived autonomy and purchase intention scales identical to those used in Study 2. Control variables were consistent with prior studies and included trust in AI recommendations (Kim et al., 2021), perceived task complexity (Bharti et al., 2026), and attitudes toward purchasing ethical products (Paul et al., 2016). Consumer knowledge was included as a moderator and assessed using a scale ($\alpha = .80$) with items such as "I can explain to other people the features of ethically produced toothpastes," consistent with prior consumer knowledge measures (Kumar et al., 2021). We also included social desirability as an additional control variable for this study. This design provides a test of H4, which predicts that consumer knowledge moderates the effect of AI nudge architecture on perceived autonomy such that the positive downstream effects of a plural AI architecture on purchase intentions are stronger among consumers with lower levels of knowledge.

Results. Participants in the plural-AI architecture condition reported significantly higher scores on the manipulation check assessing perceived AI plurality than those in the single-AI architecture condition ($M_{plural\text{-}AI} = 5.85$, $SD = 2.5$ vs. $M_{single\text{-}AI} = 2.36$, $SD = 1.68$, $t(502) = 17.93$, $p < .001$), confirming that the manipulation was perceived as intended.

To examine whether the effect of AI nudge architecture varies as a function of consumer knowledge, we first conducted a two-way ANOVA with AI condition (single vs. plural AI) and consumer knowledge (mean-centered) predicting perceived autonomy. All the relevant covariates were added to the model. The analysis revealed a significant interaction between AI condition and consumer knowledge, ($F(1, 494) = 8.5$, $p = .004$) (see Figure 2). The main effect of AI plurality was significant ($F(1, 494) = 15.15$, $p < .001$), and the main effect of knowledge was not significant ($F(1,494) = 1.5$, $p = .22$). Planned contrasts indicated that among low-knowledge consumers (using median split), participants in the AI plural systems condition reported significantly higher perceived autonomy compared to those in the single AI condition ($M_{plural\text{-}AI} = 6.33$, $SD = 1.95$; $M_{single\ AI} = 5.06$, $SD = 2.4$; $F(1, 494) = 23.5$, $p < .001$; see Figure 3). In contrast, among high-knowledge consumers, perceived autonomy did not significantly differ between the AI plural and single AI conditions ($M_{plural\text{-}AI} = 5.76$, $SD = 1.86$; $M_{single\text{-}AI} = 5.55$, $SD = 1.96$; $F(, 494) = .45$, $p = .5$). These findings suggest that AI plural systems enhance perceived autonomy primarily for consumers with lower domain knowledge.

Next, we examined purchase intention as the outcome variable. A two-way ANOVA revealed a significant interaction between AI condition and consumer knowledge, $F(1, 494) = 22.6$, $p < .001$. Planned contrasts showed that among low-knowledge consumers, purchase intention was significantly higher in the AI plural systems condition relative to the single AI condition ($M_{plural\text{-}AI} = 5.61$, $SD = 2.01$; $M_{single\text{-}AI} = 4.6$, $SD = 2.2$; $F(1, 494) = 22.6$, $p < .001$; see Figure 4). For high-knowledge consumers, purchase intention was higher in the plural AI condition than the single AI condition, however the difference was non-significant ($M_{plural\text{-}AI} = 5.61$, $SD = 2.2$; $M_{single\text{-}AI} = 5.2$, $SD = 1.8$; $F(1, 494) = 3.54$, $p = .06$).

These results provide initial evidence that the effectiveness of AI plural systems depends on consumer knowledge (H4).

Moderated Mediation

Finally, to test our hypothesized process, we conducted a moderated mediation analysis using Model 7 of PROCESS. AI nudge architecture (1 = plural system, 0 = single system) was used as the independent variable, purchase intention as the dependent variable, perceived choice autonomy as the mediator, and consumer knowledge as the moderator of the first-stage path (refer to the conceptual Figure 1). A bootstrapping analysis with 5,000 resamples was conducted to estimate conditional indirect effects. Attitude toward ethical purchase, trust in AI

recommendations, perceived task complexity, social desirability, and demographic variables were included as covariates.

Results indicate a significant interaction between AI architecture condition and consumer knowledge in predicting perceived autonomy ($\beta = -.25$, SE = .08, $p = .002$), suggesting that the effect of perceived AI plurality on perceived autonomy varies as a function of knowledge. Specifically, the AI plural system significantly increased perceived autonomy among low-knowledge consumers ($\beta = 1.34$, SE = .27, $p < .001$), whereas this effect was attenuated and non-significant among high-knowledge consumers ($\beta = .16$, SE = .26, $p = .54$).

Further analysis of conditional indirect effects revealed that the indirect effect of perceived AI plurality on purchase intention through perceived autonomy was significant at low levels of consumer knowledge (Effect = .15, BootSE = .06, 95% CI [.04, .29]) and at mean levels of knowledge (Effect = .08, BootSE = .04, 95% CI [.02, .16]), but not at high levels of knowledge (Effect = .02, BootSE = .03, 95% CI [−.04, .08]). Importantly, the index of moderated mediation was significant (Index = −.03, 95% CI [−.06, −.01]), indicating that the indirect effect of perceived AI plurality on purchase intention via perceived autonomy depends on consumer knowledge. Hence, moderated mediation (H4) is supported.

Discussion for Study 3

Results reveal that while AI plural systems function as an autonomy-supportive choice architecture, their effectiveness depends on consumers' existing knowledge. For consumers with lower domain knowledge, distributing recommendations across multiple AI agents appears to scaffold decision agency, increasing autonomy and, in turn, strengthening purchase intention. In contrast, high-knowledge consumers may rely more on internal competence than external structural cues, rendering AI plurality less influential.

General Discussion

Across three studies, this research demonstrates that the architecture of AI recommendation systems functions as a powerful nudge shaping how consumers experience choice in ethical consumption contexts. Moving beyond the informational content of recommendations, the findings show that algorithmic plurality, delivering suggestions through multiple independent AI agents rather than a single unified system, enhances consumers' sense of autonomy and strengthens their connection with recommended brands.

Study 1 provides initial evidence that breaking the single algorithmic voice through a plural AI nudge architecture increases consumers' self–brand connection with the selected ethical option. Study 2 clarifies the underlying process, showing that algorithmic plurality enhances perceived autonomy, enabling consumers to experience their choices as self-directed and personally meaningful, which in turn strengthens self–brand connection and increases purchase intentions. Study 3 further demonstrates that these autonomy-supportive effects are especially pronounced among consumers with lower domain knowledge, indicating that distributed AI recommendations can support agency when individuals feel less confident navigating complex ethical choices.

Taken together, these findings position AI recommendation architecture as a form of market-mediated nudging that can either concentrate or distribute decision authority. From a macro perspective, as algorithmic systems increasingly structure everyday consumption environments, the design of AI nudges carries societal significance. Plural AI architectures signal deliberation and balanced evaluation rather than unilateral direction, preserving consumers' autonomy while guiding ethical choices. In digitally mediated marketplaces, ensuring that AI nudges support, rather than subsume, consumer agency may be essential for sustaining consumer sovereignty and encouraging more responsible patterns of consumption.

Theoretical Implications

This research contributes to macromarketing scholarship by showing that AI recommendation architecture is not merely a platform-level interface feature but a market-system design element that can shape consumer autonomy, ethical engagement, and inclusive participation in value-based consumption. Macromarketing perspectives emphasize that market structures influence consumer welfare and societal outcomes (Layton, 2007; Mittelstaedt et al., 2006). From this perspective, AI recommendation architecture can be understood as part of the market infrastructure through which consumers encounter, evaluate, and act on ethical consumption opportunities. Extending this view, our findings show that perceived AI plurality can support consumers' sense of decision authorship in AI-mediated ethical consumption contexts. Extending this view, our findings show that plurality in AI recommendation architecture enhances perceived autonomy by distributing decision ownership back to consumers. Whereas prior research has focused on transparency, trust, and algorithmic bias in AI systems (Dietvorst et al., 2015; Logg et al., 2019; Puntoni et al., 2020), our results demonstrate that the structural design of recommendation systems, not merely the content of recommendations, shapes downstream outcomes. By enabling consumers to experience ownership over value-laden decisions, perceived AI plurality strengthens brand relationships and purchase intentions while supporting marketplace welfare. This shifts the theoretical conversation from whether AI nudges influence behavior to how AI architecture can be deliberately designed to support autonomy and responsible consumption. Building on these insights, this research advances several theoretical contributions.

First, this research advances AI nudging and digital choice architecture scholarship by identifying plurality in recommendation architecture as a critical yet overlooked lever of influence. To the best of our knowledge, this is the first study in marketing to examine how plurality in AI nudging architecture shapes perceived autonomy and downstream brand responses.

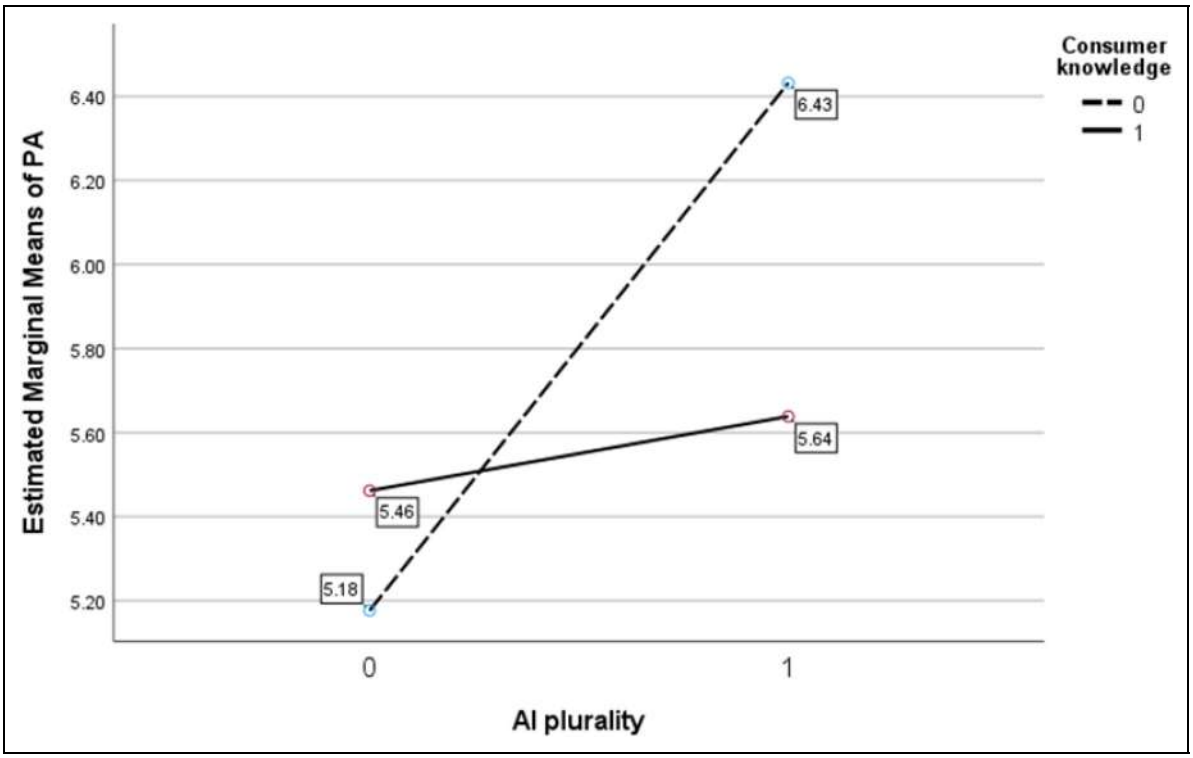


Figure 3. Interaction of AI plurality and consumer knowledge on perceived autonomy (PA); Consumer knowledge = 0 for low, 1 for high.

Prior work shows that digital interfaces and algorithmic guidance influence decisions (Adomavicius et al., 2013; Häubl & Murray, 2003), yet recommendation systems are typically treated as singular outputs and evaluated primarily through transparency. We show that presenting recommendations from multiple AI systems increases perceived autonomy, which in turn strengthens self–brand connection and purchase intentions. These findings advance research on AI-mediated decision making by demonstrating that recommendation architecture itself can foster psychological ownership and deeper consumer–brand relationships.

Second, this research contributes to marketplace governance and consumer emancipation perspectives by showing that distributing recommendation authority can restore perceived autonomy without requiring full technical transparency. Digital platforms increasingly structure demand and shape choice sets (Pasquale, 2015), and unified decision cue can reduce perceived control and trust (Dietvorst et al., 2015; Logg et al., 2019). Our findings indicate that plural recommendation structures allow consumers to experience themselves as the final decision maker rather than passive recipients of algorithmic judgment. This extends choice architecture research (Thaler & Sunstein, 2008) by demonstrating that structural pluralism can preserve perceived sovereignty and support agency in platform-mediated markets.

Third, this research advances sustainability and ethical consumption theory by identifying perceived autonomy as a mechanism through which consumers engage more meaningfully with ethical decisions. Sustainable options are often perceived as involving sacrifice (Luchs et al. 2010; White et al., 2019), yet ethical consumption can provide moral meaning and reinforce identity expression (Griskevicius et al., 2010; Reed et al., 2012). We demonstrate that when consumers experience responsibility and authorship in resolving ethical considerations, decisions are experienced as self-endorsed rather than externally imposed, strengthening identity-relevant brand connections and purchase intentions. The results suggest that perceived AI plurality enhances marketplace welfare by enabling consumers to experience ownership over value-based decisions, thereby strengthening brand relationships and purchase intentions. This shifts the theoretical conversation from whether AI nudges influence behavior to how AI architectures can be deliberately designed to support autonomy and ethical consumption simultaneously.

Fourth, this research extends consumer identity and marketplace culture scholarship by showing that perceived authorship in AI-mediated decisions strengthens self–brand connection. Consumption is a central site of identity construction (Belk, 1988; Arnould & Thompson, 2005), and self–brand connections deepen when choices reflect the self (Escalas & Bettman, 2005). Our findings indicate that identity expression depends not only on what consumers choose but also on whether they feel autonomy in the decision process, highlighting how AI system design shapes relational outcomes in value-laden contexts.

Finally, this research contributes to macromarketing discussions of consumer capability and marketplace inclusion

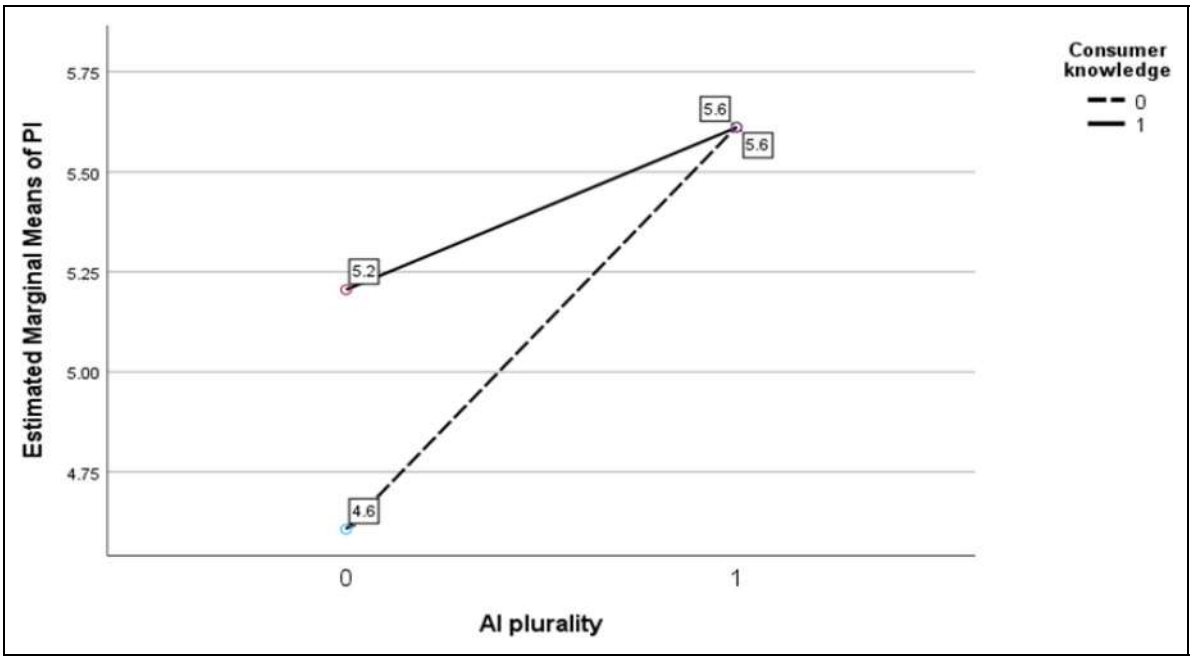


Figure 4. Interaction of AI plurality and consumer knowledge on purchase intention (PI); Consumer knowledge = 0 for low, 1 for high.

by identifying consumer knowledge as a boundary condition in AI-mediated choice. Consumer knowledge shapes reliance on decision aids (Alba & Hutchinson, 1987; Bettman & Park, 1980), and low knowledge can deter green consumption (Lee et al., 2020; Luo et al., 2023; Roh et al., 2022). We find that perceived AI plurality particularly benefits less knowledgeable consumers by functioning as a compensatory mechanism. It provides structured guidance that supports ethical decision-making while preserving perceived autonomy. In doing so, perceived AI plurality may enable more inclusive participation in value-based consumption among consumers who might otherwise feel less equipped to evaluate ethical marketplace options. In this way, perceived AI plurality highlights the emancipatory potential of marketplace design: markets can constrain or enable agency depending on how they are structured (Mittelstaedt et al., 2006), and AI architecture can serve as a societal lever shaping empowerment, ethical engagement, and well-being in platform-mediated markets (Cohen, 2019).

Managerial and Public Policy Implications

The findings suggest that AI recommendation systems should be designed and governed in ways that preserve consumer agency while advancing relational, ethical, and societal outcomes. Presenting a single AI recommendation can signal decisional finality and reduce perceived control (Dietvorst et al., 2015). In contrast, presenting recommendations from multiple AI systems enhances perceived autonomy and allows consumers to experience decisions as self-endorsed, reinforcing perceptions of fairness and consumer sovereignty (Thaler & Sunstein, 2008). Because perceived authorship over value-laden choices strengthens relational meaning, autonomy-supportive AI design can potentially foster deeper consumer–brand connections, trust, and long-term loyalty (Escalas & Bettman, 2005; Morgan & Hunt, 1994). Rather than positioning AI as a definitive decision maker, firms can design recommendation environments that support consumer deliberation while preserving efficiency. One practical approach is the use of panel-based AI interfaces, in which recommendations are presented through multiple AI agents rather than a single AI source. Such interfaces can preserve the efficiency benefits of AI guidance while making the recommendation environment feel less singular and more autonomy-supportive.

The findings also offer guidance for promoting sustainable consumption. Sustainability initiatives often fail when framed as sacrifice (White et al., 2019). Presenting multiple sustainability-oriented options and encouraging consumer deliberation can transform ethical considerations into identity-affirming decisions, increasing willingness to support value-consistent choices (Kilbourne & Pickett, 2008; Reed et al., 2012). In this way, firms can support ethical consumption without imposing prescriptive directives. Plural recommendation structures appear particularly beneficial for less knowledgeable consumers, suggesting that autonomy-enhancing design may improve decision confidence and reduce informational inequities. Such designs support broader marketplace participation and align with macromarketing goals of enhancing consumer capability and inclusion (Ozanne & Murray, 1995; Shultz & Holbrook, 1999).

From a public policy perspective, the findings indicate that protecting consumer autonomy in algorithmically mediated markets requires more than transparency and disclosure. Empowerment-oriented design that preserves perceived agency

may be essential for maintaining consumer sovereignty in digital marketplaces (Cohen, 2019). Regulators and standards bodies may therefore consider encouraging autonomy-supportive recommendation practices in high-stakes domains such as sustainability, health, and financial decision-making, where preserving agency is critical. Because recommendation structures may disproportionately influence less knowledgeable consumers, policymakers should also evaluate equity implications and promote standards that support inclusive, autonomy-enhancing decision environments (Ozanne & Murray, 1995).

Taken together, these findings suggest that AI-mediated marketplaces can either constrain or enable consumer emancipation. Designing recommendation systems that support perceived autonomy allows markets to guide decisions while preserving agency, enabling consumers to act in ways that reflect their values and potentially contribute to broader societal well-being.

Limitations and Future Research Directions

This research advances understanding of how plurality in AI nudging shapes ethical consumption decisions. At the same time, several limitations point to important avenues for future inquiry, particularly from a macromarketing perspective concerned with marketplace structures, consumer welfare, and the societal consequences of algorithmic design.

First, perceived AI plurality was operationalized using three AI-generated recommendations. This design allows consumers to perceive meaningful plurality without excessive cognitive burden; however, the optimal scale of plurality remains unknown. Too few AI voices may fail to signal genuine diversity in recommendation logic, whereas too many may introduce cognitive overload, decision fatigue, or diffusion of responsibility. Future research should examine how varying degrees of plurality influence decision quality and perceived empowerment, and whether these effects follow nonlinear patterns. Understanding the threshold at which plurality shifts from empowering to overwhelming is essential for designing recommendation environments that support consumer welfare.

Second, the findings are based on participants drawn from a limited cultural context. Cultural norms shape perceptions of authority, autonomy, and decision responsibility. Research suggests that reliance on algorithmic advice and reactions to automated decision aids vary across social and institutional environments (Logg et al., 2019; Puntoni et al., 2020). In contexts characterized by higher power distance or stronger reliance on expert authority, unified AI recommendations may be preferred, whereas consumers in more individualistic contexts may respond more positively to plural recommendation architectures. Cross-cultural research is therefore needed to clarify how AI-mediated choice environments interact with cultural values and governance traditions.

Furthermore, future research could examine whether the effects of perceived AI plurality extend to convergent multi-agent displays, in which multiple AI agents recommend the same product or set of products. Such work would help distinguish plurality as a source-architecture cue from consensus as a signal of agreement or certainty. This distinction would clarify when multiple AI agents enhance autonomy and decision authorship, and when they operate primarily through perceived validation or endorsement strength.

Third, this research focuses on ethical consumption decisions. Future work should examine whether plurality effects extend to other domains, including financial planning, health decisions, and public service contexts, where perceived risk, accountability expectations, and expertise requirements differ. Recommendation systems are increasingly embedded across institutional domains, and their structural design may shape how responsibility and trust are distributed between consumers and automated systems (Puntoni et al., 2020). Investigating high-stakes contexts would extend macromarketing understanding of how AI-mediated infrastructures influence societal decision-making.

Fourth, while this research conceptualizes plurality as a structural design feature, it does not manipulate transparency regarding how recommendations are generated. Prior research shows that algorithmic presentation and recommendation framing can influence consumer preferences and decision processes (Adomavicius et al., 2013). Future research should examine how plurality interacts with explainability, disclosure, and governance mechanisms to shape perceptions of legitimacy, accountability, and trust. Such inquiry is particularly relevant in light of ongoing regulatory debates surrounding algorithmic transparency and responsible AI deployment. Relatedly, future research could examine perceived AI plurality in repeated-interaction contexts, where consumers develop trust in specific AI agents through accumulated platform experience, personalization history, and demonstrated prediction accuracy over time.

Finally, although the present findings highlight the autonomy-enhancing potential of AI plurality, future research should examine potential unintended consequences. Plural recommendations may increase decision effort, create perceived responsibility burdens, or shift accountability from firms to consumers. While the present studies controlled for choice complexity, real-world marketplaces often involve higher information density and competing claims, conditions under which plurality may amplify cognitive burden rather than empowerment. From a macromarketing perspective, understanding when plurality enhances consumer agency versus when it transfers decision burdens is critical for evaluating the societal welfare implications of AI-mediated marketplaces.

Taken together, these limitations underscore the importance of examining how algorithmic design choices influence consumer autonomy, marketplace fairness, and the governance of AI-enabled consumption systems.

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Supplemental Material

Supplemental material for this article is available online.

References

Adomavicius, G., Bockstedt, J. C., Curley, S. P., & Zhang, J. (2013). Do recommender systems manipulate consumer preferences? A study of anchoring effects. *Information Systems Research*, *24*(4), 956–975. https://psycnet.apa.org/doi/10.1287/isre.2013.0497 https://doi.org/10.1287/isre.2013.0497

Adomavicius, G., & Tuzhilin, A. (2005). Toward the next generation of recommender systems: A survey of the state-of-the-art and possible extensions. *IEEE Transactions on Knowledge and Data Engineering*, *17*(6), 734–749. https://doi.org/10.1109/TKDE.2005.99

Alba, J. W., & Hutchinson, J. W. (1987). Dimensions of consumer expertise. *Journal of Consumer Research*, *13*(4), 411–454. https://doi.org/10.1086/209080

André, Q., Carmon, Z., Wertenbroch, K., Crum, A., Frank, D., Goldstein, W., Huber, J., Van Boven, L., Weber, B., & Yang, H. (2018). Consumer choice and autonomy in the age of artificial intelligence and big data. *Customer Needs and Solutions*, *5*(1–2), 28–37. https://doi.org/10.1007/s40547-017-0085-8

Arnould, E. J., & Thompson, C. J. (2005). Consumer Culture Theory (CCT): Twenty years of research. *Journal of Consumer Research*, *31*(4), 868–882. https://doi.org/10.1086/426626

Auh, S., Menguc, B., Fisher, M., & Haddad, A. (2011). The perceived autonomy–perceived service climate relationship: The contingency effect of store-level tenure diversity. *Journal of Retailing and Consumer Services*, *18*(6), 509–520. https://doi.org/10.1016/j.jretconser.2011.07.001

Bandura, A. (2001). Social cognitive theory: An agentic perspective. *Annual Review Psychology*, *52*, 1–26. https://doi.org/10.1146/annurev.psych.52.1.1

Belk, R. W. (1988). Possessions and the extended self. *Journal of Consumer Research*, *15*(2), 139–168. https://doi.org/10.1086/209154

Bettman, J. R., & Park, C. W. (1980). Effects of prior knowledge and experience and phase of the choice process on consumer decision processes: A protocol analysis. *Journal of Consumer Research*, *7*(3), 234–248. https://doi.org/10.1086/208812

Bhardwaj, E., Gujral, H., Wu, S., Zogheib, C., Maharaj, T., & Becker, C. (2024). Machine learning data practices through a data curation lens: An evaluation framework. In *Proceedings of the 2024 ACM conference on fairness, accountability, and transparency (FAccT '24)* (pp. 1055–1067). ACM. https://doi.org/10.1145/3630106.3658955

Bharti, M., & Bharti, M. (2025). Flipping the script: Harnessing consumer knowledge, message order and refutation for effective two-sided green advertising. *Psychology & Marketing*, *42*(5), 1361–1378. https://doi.org/10.1002/mar.22184

Bharti, M., Bharti, M., Ahmad Wani, T., & Shahid, S. (2026). The FOMO effect: How AI recommendations drive consumer dependence and weaken brand-self connections. *Journal of Consumer Behaviour*, *25*(3), 1–29. https://doi.org/10.1002/cb.70117

Carrington, M. J., Neville, B. A., & Whitwell, G. J. (2014). Lost in translation: Exploring the ethical consumer intention–behavior gap. *Journal of Business Research*, *67*(1), 2759–2767. https://doi.org/10.1016/j.jbusres.2012.09.022

Casais, B., & Faria, J. (2021). The intention-behavior gap in ethical consumption: Mediators, moderators and consumer profiles based on ethical priorities. *Journal of Macromarketing*, *42*(1), 100–113. https://doi.org/10.1177/02761467211054836

Cheng, S. Y., White, T. B., & Chaplin, L. N. (2011). The effects of self-brand connections on responses to brand failure: A new look at the consumer–brand relationship. *Journal of Consumer Psychology*, *22*(2), 280–288. https://doi.org/10.1016/j.jcps.2011.05.005

Chernev, A., Hamilton, R., & Gal, D. (2011). Competing for consumer identity: Limits to self-expression and the perils of lifestyle branding. *Journal of Marketing*, *75*(3), 66–82. https://doi.org/10.1509/jmkg.75.3.66

Cho, E., & Rha, J. (2026). Does consumer autonomy matter? Consumer perceptions of AI-driven recommendations and the mediating role of consumer autonomy in consumer satisfaction. *International Journal of Human-Computer Interaction*, *42*(14), 10949–10971. https://doi.org/10.1080/10447318.2025.2587927

Choi, S., & Yi, Y. (2025). The impact of service agent type on satisfaction in green purchasing: A comparative study of AI and human agents. *Journal of Retailing and Consumer Services*, *87*, 1–10. https://doi.org/10.1016/j.jretconser.2025.104355

Cohen, J. E. (2019). *Between truth and power: The legal constructions of informational capitalism*. Oxford University Press.

Deci, E. L., & Ryan, R. M. (2000). The "what" and "why" of goal pursuits: Human needs and the self-determination of behavior. *Psychological Inquiry*, *11*(4), 227–268. https://doi.org/10.1207/S15327965PLI1104_01

Dietvorst, B. J., Simmons, J. P., & Massey, C. (2015). Algorithm aversion: People erroneously avoid algorithms after seeing them err. *Journal of Experimental Psychology: General*, *144*(1), 114–126. https://doi.org/10.1037/xge0000033

Escalas, J. E. (2004). *Self-Brand Connection scale* [Dataset]. In PsycTESTS Dataset. https://doi.org/10.1037/t15286-000

Escalas, J. E., & Bettman, J. R. (2003). You are what they eat: The influence of reference groups on Consumers' connections to brands. *Journal of Consumer Psychology*, *13*(3), 339–348. https://doi.org/10.1207/S15327663JCP1303_14

Escalas, J. E., & Bettman, J. R. (2005). Self-Construal, reference groups, and brand meaning. *Journal of Consumer Research*, *32*(3), 378–389. https://doi.org/10.1086/497549

Escalas, J. E., & Bettman, J. R. (2017). Connecting with celebrities: How consumers appropriate celebrity meanings for a sense of belonging. *Journal of Advertising*, *46*(2), 297–308. https://doi.org/10.1080/00913367.2016.1274925

Gkikas, D., & Theodoridis, P. (2021). AI in consumer behavior. In M. Virvou, G. A. Tsihrintzis, L. H. Tsoukalas, & L. C. Jain (Eds.), *Advances in artificial intelligence-based technologies. Learning and analytics in intelligent systems*, (Vol. 22, pp. 147–176). Springer. https://doi.org/10.1007/978-3-030-80571-5_10

Grewal, D., Krishnan, R., Baker, J., & Borin, N. (1998). The effect of store name, brand name, and price discounts on consumers' evaluations and purchase intentions. *Journal of Retailing*, *74*(3), 331–352. https://doi.org/10.1016/S0022-4359(99)80099-2

Griskevicius, V., Tybur, J. M., & Van den Bergh, B. (2010). Going green to be seen: Status, reputation, and conspicuous conservation. *Journal of Personality and Social Psychology*, *98*(3), 392–404. https://doi.org/10.1037/a0017346

Häubl, G., & Murray, K. B. (2003). Preference construction and persistence in digital marketplaces: The role of electronic recommendation agents. *Journal of Consumer Psychology*, *13*(1–2), 75–91. https://doi.org/10.2139/ssrn.964192

Häubl, G., & Trifts, V. (2000). Consumer decision making in online shopping environments: The effects of interactive decision aids. *Marketing Science*, *19*(1), 4–21. https://doi.org/10.1287/mksc.19.1.4.15178

Hermann, E. (2022). Leveraging artificial intelligence in marketing for social good—an ethical perspective. *Journal of Business Ethics*, *179*(1), 43–61. https://doi.org/10.1007/s10551-021-04843-y

Homburg, C., Koschate, N., & Hoyer, W. D. (2005). Do satisfied customers really pay more? A study of the relationship between customer satisfaction and willingness to pay. *Journal of Marketing*, *69*(2), 84–96. https://doi.org/10.1509/jmkg.69.2.84.60760

Hyman, M. R., Kostyk, A., & Trafimow, D. (2022). True consumer autonomy: A formalization and implications. *Journal of Business Ethics*, *183*(3), 841–863. https://doi.org/10.1007/s10551-022-05114-0

Iyengar, S. S., & Lepper, M. R. (2000). When choice is demotivating: Can one desire too much of a good thing? *Journal of Personality and Social Psychology*, *79*(6), 995–1006. https://doi.org/10.1037/0022-3514.79.6.995

Johnson, E. J., Shu, S. B., Dellaert, B. G. C., Fox, C., Goldstein, D. G., Häubl, G., Larrick, R. P., Payne, J. W., Peters, E., Schkade, D., Wansink, B., & Weber, E. U. (2012). Beyond nudges: Tools of a choice architecture. *Marketing Letters*, *23*(2), 487–504. https://doi.org/10.1007/s11002-012-9186-1

Kilbourne, W., & Pickett, G. (2008). How materialism affects environmental beliefs, concern, and environmentally responsible behavior. *Journal of Business Research*, *61*(9), 885–893. https://doi.org/10.1016/j.jbusres.2007.09.016

Kim, J., Giroux, M., & Lee, J. C. (2021). When do you trust AI? The effect of number presentation detail on consumer trust and acceptance of AI recommendations. *Psychology & Marketing*, *38*(7), 1140–1155. https://doi.org/10.1002/mar.21498

Kumar, P., Polonsky, M., Dwivedi, Y. K., & Kar, A. (2021). Green information quality and green brand evaluation: The moderating effects of eco-label credibility and consumer knowledge. *European Journal of Marketing*, *55*(7), 2037–2071.

Kumari, P., & Raj, A. (2024). The Role of AI Nudges in Optimizing Consumer Behavior in Financial and E-Commerce Contexts. https://dx.https://doi.org/10.2139/ssrn.4937811

Layton, R. A. (2007). Marketing systems—A core macromarketing concept. *Journal of Macromarketing*, *27*(3), 227–242. https://doi.org/10.1177/0276146707302836

Lee, E. J., Bae, J., & Kim, K. H. (2020). The effect of environmental cues on the purchase intention of sustainable products. *Journal of Business Research*, *120*, 425–433. https://doi.org/10.1016/j.jbusres.2019.10.048

Logg, J. M., Minson, J. A., & Moore, D. A. (2019). Algorithm appreciation: People prefer algorithmic to human judgment. *Organizational Behavior and Human Decision Processes*, *151*, 90–103. https://doi.org/10.1016/j.obhdp.2018.12.005

Luchs, M. G., Naylor, R. W., Irwin, J. R., & Raghunathan, R. (2010). The sustainability liability: Potential negative effects of ethicality on product preference. *Journal of Marketing*, *74*(5), 18–31. https://doi.org/10.1509/jmkg.74.5.18

Luo, G., Zheng, H., & Guo, Y. (2023). Impact of consumer information capability on green consumption intention: The role of green trust and media publicity. *Frontiers in Psychology*, *14*, 1247479. https://doi.org/10.3389/fpsyg.2023.1247479

Luo, Y., Kumar, N., & Yazdanmehr, A. (2026). AI nudging and decision quality: Evidence from randomized experiments in online recommendation setting. *Decision Support Systems*, *200*, 114565. https://doi.org/10.1016/j.dss.2025.114565

MacKenzie, I., Meyer, C., & Noble, S. (2013). *How retailers can keep up with consumers*. McKinsey & Company. https://www.mckinsey.com/industries/retail/our-insights/how-retailers-can-keep-up-with-consumers.

McCracken, G. (1986). Culture and consumption: A theoretical account of the structure and movement of the cultural meaning of consumer goods. *Journal of Consumer Research*, *13*(1), 71–84. https://doi.org/10.1086/209048

Mittelstaedt, J. D., Kilbourne, W. E., & Mittelstaedt, R. A. (2006). Macromarketing as agorology: Macromarketing theory and the study of the agora. *Journal of Macromarketing*, *26*(2), 131–142. https://doi.org/10.1177/0276146706290921

Moore, D. J., & Homer, P. M. (2008). Self-brand connections: The role of attitude strength and autobiographical memory primes. *Journal of Business Research*, *61*(7), 707–714. https://doi.org/10.1016/j.jbusres.2007.09.002

Morgan, R. M., & Hunt, S. D. (1994). The commitment-trust theory of relationship marketing. *Journal of Marketing*, *58*(3), 20–38. https://doi.org/10.1177/002224299405800302

Necula, S.-C., & Păvăloaia, V.-D. (2023). AI-Driven Recommendations: A systematic review of the state of the art in E-commerce. *Applied Sciences*, *13*(9), 5531. https://doi.org/10.3390/app13095531

Nokhiz, P., & Ruwanpathirana, A. K. (2025). Consumer autonomy or illusion? Rethinking consumer agency in the age of algorithms.

Journal of Social Computing, *6*(3), 184–208. https://doi.org/10.23919/JSC.2025.0015

Oxenfeldt, A. R. (1950). Consumer knowledge: Its measurement and extent. *The Review of Economics and Statistics*, *32*(4), 300. https://doi.org/10.2307/1925578

Ozanne, J. L., & Murray, J. B. (1995). Uniting critical theory and public policy to create the reflexively defiant consumer. *American Behavioral Scientist*, *38*(4), 516–525. https://doi.org/10.1177/0002764295038004003

Pasquale, F. (2015). *The black box society: The secret algorithms that control money and information*. Harvard University Press.

Paul, J., Modi, A., & Patel, J. (2016). Predicting green product consumption using theory of planned behavior and reasoned action. *Journal of Retailing and Consumer Services*, *29*, 123–134. https://doi.org/10.1016/j.jretconser.2015.11.006

Pillai, K. G., Brusco, M., Goldsmith, R., & Hofacker, C. (2015). Consumer knowledge discrimination. *European Journal of Marketing*, *49*(1/2), 82–100. https://doi.org/10.1108/ejm-05-2012-0288

Puntoni, S., Reczek, R. W., Giesler, M., & Botti, S. (2020). Consumers and artificial intelligence: An experiential perspective. *Journal of Marketing*, *84*(1), 131–151. https://doi.org/10.1177/0022242920953847

Qadri, U. A., Ahmed Moustafa, A. M., & Waqas, M. (2026). When and how AI personalization drives sustainable purchases: The roles of relevance, privacy, and transparency in eco-friendly advertising. *Journal of Retailing and Consumer Services*, *89*, 104592. https://doi.org/10.1016/j.jretconser.2025.104592

Ratchford, B. T. (2001). The economics of consumer knowledge: Table 1. *Journal of Consumer Research*, *27*(4), 397–411. https://doi.org/10.1086/319617

Reed, A., Forehand, M. R., Puntoni, S., & Warlop, L. (2012). Identity-based consumer behavior. *International Journal of Research in Marketing*, *29*(4), 310–321. https://doi/10.1016/j.ijresmar.2012.08.002 https://doi.org/10.1016/j.ijresmar.2012.08.002

Roh, T., Seok, J., & Kim, Y. (2022). Unveiling ways to reach organic purchase: Green perceived value, perceived knowledge, attitude, subjective norm, and trust. *Journal of Retailing and Consumer Services*, *67*, 102988. https://doi.org/10.1016/j.jretconser.2022.102988

Schwarz, O. (2025). The post-choice society: Algorithmic prediction and the decentring of choice. *Theory, Culture & Society*, *42*(4), 3–22. https://doi.org/10.1177/02632764251322062 (Original work published 2025). https://doi.org/10.1177/02632764251322062

Selinger, E., & Whyte, K. P. (2011). Is there a right way to nudge? The practice and ethics of choice architecture. *Sociology Compass*, *5*(10), 923–935. https://doi.org/10.1111/j.1751-9020.2011.00413.x

Shultz, C. J., & Holbrook, M. B. (1999). Marketing and the tragedy of the commons: A synthesis, commentary, and analysis for action. *Journal of Public Policy & Marketing*, *18*(2), 218–229. https://doi.org/10.1177/074391569901800208

Smith, B., & Linden, G. (2017). Two decades of recommender systems at Amazon.com. *IEEE Internet Computing*, *21*(3), 12–18. https://doi.org/10.1109/MIC.2017.72

Statista. (2024). Shoppers using generative AI tools for recommendations in 2023, by generation. Statista. https://www.statista.com/statistics/1380351/gen-ai-for-product-recommendations-by-generation/

Thaler, R. H., & Sunstein, C. (2008). *Nudge - Improving Decisions About Health, Wealth and Happiness*. Penguin Books.

Thaler, R. H., Sunstein, C. R., & Balz, J. P. (2012). Choice architecture. In E. Shafir (Ed.), *The behavioral foundations of public policy* (pp. 428–439). Princeton University Press.

Wan, J., & Bharti, M. (2025). From Price to Quantity: Redefining How Consumers Pay the Ethical Premium. *Journal of Business Ethics*. Advance online publication. https://doi.org/10.1007/s10551-025-06178-4

Weinmann, M., Schneider, C., & vom Brocke, J. (2016). Digital nudging. *Business & Information Systems Engineering*, *58*(6), 433–436. https://doi.org/10.1007/s12599-016-0453-1

White, K., & Dahl, D. W. (2007). Are all out-groups created equal? Consumer identity and dissociative influence. *Journal of Consumer Research*, *34*(4), 525–536. https://doi.org/10.1086/520077

White, K., Habib, R., & Hardisty, D. J. (2019). How to SHIFT consumer behaviors to be more sustainable: A literature review and guiding framework. *Journal of Marketing*, *83*(3), 22–49. https://doi.org/10.1177/0022242919825649

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